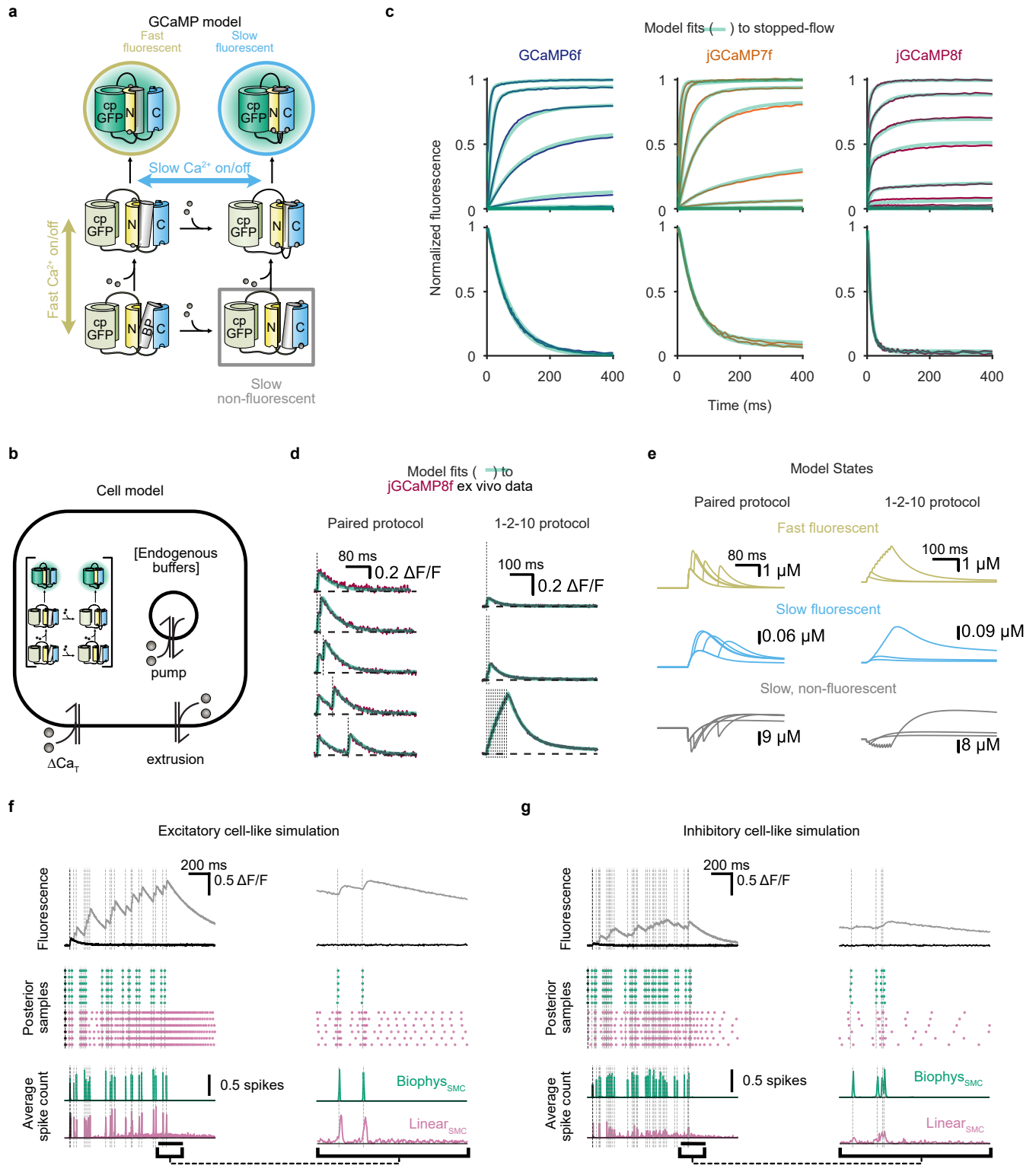

a
GCaMP model
Fast fluorescent
Slow fluorescent
cp GFP
N
C
Slow Ca2+ on/off
Fast Ca2+ on/off
BP
Slow non-fluorescent
c
Model fits ( ) to stopped-flow
GCaMP6f
jGCaMP7f
jGCaMP8f
Normalized fluorescence
Time (ms)
b
Cell model
[Endogenous buffers]
pump
ΔCaT
extrusion
d
Model fits ( ) to jGCaMP8f ex vivo data
Paired protocol
1-2-10 protocol
80 ms
0.2 ΔF/F
100 ms
e
Model States
Paired protocol
1-2-10 protocol
Fast fluorescent
1 μM
Slow fluorescent
0.06 μM
0.09 μM
Slow, non-fluorescent
9 μM
8 μM
f
Excitatory cell-like simulation
200 ms
0.5 ΔF/F
Fluorescence
Posterior samples
Average spike count
0.5 spikes
BiophysSMC
LinearSMC
g
Inhibitory cell-like simulation